LiveKit Voice Agent → Datadog Agent Observability

Making LLM calls and tool calls visible

The problem

A LiveKit Agents worker runs each call (speech → LLM → tools → speech) in its own process. By default none of that reaches Datadog — the **LLM turns and tool calls were invisible**, so you couldn't see what the model decided or which tools it invoked.

What we did

1. Enable LLM Observability in the worker

ddtrace's OpenAI integration then auto-captures every model call as an **LLM span** with the prompt, reply, **token counts**, and the **tool_calls the model requested** — no code for the LLM turns.

```
DD_LLMOBS_ENABLED=1
DD_LLMOBS_ML_APP=livekit-loan-agent
# start with:  ddtrace-run python agent.py start
```

2. Run calls in-process (the critical fix)

LiveKit normally forks a **subprocess** per call, whose telemetry never exported. THREAD execution keeps the call in the instrumented process, so spans actually flush to Datadog.

```
cli.run_app(WorkerOptions(entrypoint_fnc=entrypoint,
                          job_executor_type=JobExecutorType.THREAD))
```

3. Instrument the tools & hold one trace per call

Wrapping each tool in **LLMObs.tool(...)** makes the tool call a first-class span with its arguments and result. A held-open **voice_agent_call** root (re-anchored on each step) groups the whole call into one agent trace.

```
root = LLMObs.workflow(name="voice_agent_call", session_id=room)

@function_tool
async def check_loan_eligibility(self, ctx, annual_income, requested_amount=0):
    self._anchor()                      # nest under the call root
    with LLMObs.tool(name="check_loan_eligibility") as span:
        LLMObs.annotate(span=span, input_data={"annual_income": annual_income})
        result = {"approved": True, "apr": 6.5, "amount": 18000}
        LLMObs.annotate(span=span, output_data=result)
        return result
```

4. Do NOT set DD_TRACE_OTEL_ENABLED

With it on, Datadog reads spans via the OpenTelemetry GenAI convention and expects gen_ai.operation.name, mislabelling every span as a generic "workflow". Leaving it off keeps spans classified natively as llm / tool / workflow.

The result — one real call

Caller: *“I’d like a loan for a car repair, I make about $60,000 a year… yes, let’s proceed.”* In LLM Observability that call is **one trace**:

Every **LLM turn** shows its tokens and the tool_calls it requested; every **tool span** shows its arguments and result — all under one call, grouped by session. Where to look: **LLM Observability** → ml_app livekit-loan-agent (Sessions view for the full conversation).

In one line

Enable LLM Obs (**DD_LLMOBS_ENABLED** + **ddtrace-run**) → run calls in-thread so they export → wrap the tools in **LLMObs.tool** under a held-open **voice_agent_call** root → keep **DD_TRACE_OTEL_ENABLED off** so spans classify natively.

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